

C# fundamentals

Introduction to .NET Full stack development

ASP Core

Prerequisites:

- **C#**
- **OOP**
- **Principles**
- **SQL Server**
- **LINQ**
- **EF**

ASP Core

What's ASP. Net Core?

- Asp.net is a popular web- development framework for building we apps on the .Net platform.
- Asp.net Core is the open source version of asp.net, that runs on macOS, Linux, and Windows.
- Asp.Net Core was first released in 2016.

Why .NET Core?

Cross Platform:

Easier for developers to create applications that can be used on multiple platforms.

Performance:

.NET Core is built with performance in mind.

Cloud-ready:

.NET Core is designed to work well in cloud environments.

Modern development:

.NET Core provides developers with modern development tools and techniques.

ASP Core

What's API?

API is acronym for Application Programming Interface which is a mechanisms that enable two software components to communicate with each other using a set of definitions and protocols.

APIs Should:

Be stateless

Respect HTTP methods.

Respect HTTP standard status codes.

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- في سياق الويب تمثل ويب سيرفيس

ASP Core

What's HTTP?

- HTTP is acronym for Hypertext Transfer Protocol which is the protocol that enables the transfer for data over internet, allowing users to access websites and other online resources.



HTTP Methods/Verbs

Method	Usage
GET	Used to get data from API
POST	Used to create new records
PUT	Used to update existing records
DELETE	Used to delete records

HTTP Status Codes

Method	Usage
200-299	Successful response
300-399	Redirection
400-499	Client errors
500-599	Server errors

Most Common HTTP Status Codes

Code	Name	Usage
200	OK	Successful response
201	Created	Resource has been created
204	No Content	Request succeeded but no content returned from the server
400	Bad Request	Request is not processed because it's not valid
401	Unauthorized	The client is not <u>authenticated</u>
403	Forbidden	The client is authenticated but doesn't have permission to perform the request
404	Not Found	The requested resource was not found
405	Method Not Allowed	The request method is not supported by the target resource
409	Conflict	The request conflicts with the current state

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What's JSON?

JSON is acronym for JavaScript Object Notation, is open standard format, which is lightweight and text-based, designed explicitly for human-readable data interchange.

```
public class Person
{
    public int Id { get; set; }
    public string Name { get; set; }
    public DateTime BirthDate { get; set; }
    public Address Address { get; set; }
}

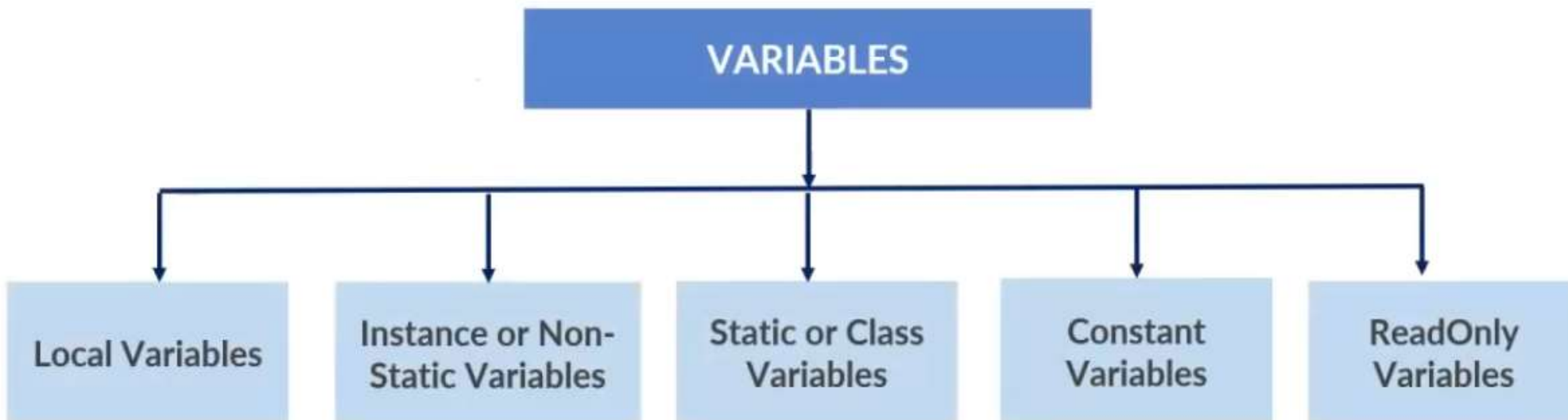
public class Address
{
    public string City { get; set; }
    public string Area { get; set; }
    public string BuildingNumber { get; set; }
    public string ApartmentNumber { get; set; }
}
```

```
{
  "Person": {
    "Id": 0,
    "Name": "string",
    "BirthDate": "1/1/0001 12:00:00 AM",
    "Address": {
      "City": "string",
      "Area": "string",
      "BuildingNumber": "string",
      "ApartmentNumber": "string"
    }
  },
  "Address": {
    "City": "string",
    "Area": "string",
    "BuildingNumber": "string",
    "ApartmentNumber": "string"
  }
}
```


Installing visual studio

- In order to code in C#, we need to install the latest stable release of Microsoft Visual Studio.
- Which is an integrated development environment (IDE) from Microsoft.
- First, visit the following Visual Studio free download link.
 - <https://visualstudio.microsoft.com/downloads/>
- It used to code in C# and develop computer programs, websites, web apps, mobile apps, etc.

TYPES OF VARIABLES



// Expression Lambda:

```
var numbers = new int[] { 2, 5, 6, 5, 1, 3, 5, 7 };  
var count = numbers.Count(x => x == 5);  
Console.WriteLine(count);
```

// Statement Lambda

```
List<int> numbers2 = new List<int> { 2, 5, 6, 5, 1, 3, 5, 7 };  
count = numbers.Count(x => { return x == 5; });  
Console.WriteLine(count);
```